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Location of Energy Barriers. I. Effect on the Dynamics of Reactions $A+BC$

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The dynamics of exchange reactions $A+BC \rightarrow AB+C$ have been examined on two types of potential-energy hypersurfaces that differed in the location of the energy barrier along the reaction coordinate. On "surface I" the barrier was in the entry valley of the energy surface, along the approach coordinate. On "surface II" the barrier was in the exit valley of the energy surface, along the retreat coordinate. The classical barrier height was $E_c = 7.0$ kcal mole⁻¹ on both surfaces, and was displaced from the corner of the energy surface by the same amount; on surface I, $r_1^\ddagger = 1.20$ Å, $r_2^\ddagger = 0.80$ Å; on surface II, $r_1^\ddagger = 0.80$ Å, $r_2^\ddagger = 1.20$ Å ($r_1 = r_{AB}$, $r_2 = r_{BC}$, and the superscript $\ddagger$ refers to the location of the crest of the barrier). Three-dimensional (3D) classical trajectory calculations were performed for the mass combination $m_A = m_B = m_C$ at several reagent energies. The reagent energy took the form of translation, vibration or an equilibrium distribution of the two. The main findings were that translation was markedly more effective than vibration in promoting reaction on surface I, and vibration markedly more effective than translation in promoting reaction on surface II. The total reactive cross section with the entire reagent energy vested in translation was symbolized S^T , with the reagent energy (but for 1.5 kcal) in vibration, S^V , and with an equilibrium distribution over reagent translation and vibration, S^{eq} . On surface I $S^T \gg S^V$; on surface II $S^V \gg S^T$. Close to the threshold for S^T on surface I, $S^T/S^{eq} \sim 10$; close to the threshold for S^V , on surface II, $S^V/S^{eq} \sim 10$. At high reagent energies ($2 \times$ threshold) on surface I S^T/S^{eq} fell to 2, whereas on surface II S^V/S^{eq} increased to extremely large values. Product energy and angular distributions were recorded for two reagent energies. On surface I with low translational energy in the reagents a major part of the available energy appeared as vibration in the molecular product. At higher collision energy this fraction decreased. On surface II with low vibrational energy in the reagents only a small part of the available energy appeared as vibration in the product. At higher vibrational energy this fraction increased. The product angular distribution at low reagent translational energy on surfaces I and II corresponded to backward-peaked scattering of the molecular product. At increased reagent energy on both surfaces the distribution shifted forward (this is a novel phenomenon in the case of increased reagent vibration; surface II).

I. INTRODUCTION

Starting with the work of Wall, Hiller, and Mazur¹ a considerable number of classical trajectory studies have been made of the dynamics of simple three-body exchange reactions, $A+BC \rightarrow AB+C$. Broadly speaking, two different types of study can be distinguished: (1) those in which the prime objective was to provide a model for a particular reactive system, and (2) those in which the prime objective was the examination of the effect of systematic variation in some of the parameters governing the reaction dynamics. In practice the distinction is not at all clear cut, and studies which could be placed in one category provide important information in the other.

Work that might be placed in category (1) is the study of the systems $K+ICH_3$ (treated as $A+BC$),² $H+H_2$,³ and $K+Br_2$.⁴ In category (2) there have been studies of the effect of systematic variation in the proportion of "attraction" to "repulsion" on London-Eyring-Polanyi-Sato (LEPS) energy surfaces,^{5,6} the

effect of systematic variation in reactant masses on these same surfaces,^{6,7} and the effect of systematic variation in electron-jump distance, product repulsion, and charge migration ($B-C \rightarrow BC^-$) on electron-jump surfaces ($A+BC \rightarrow A^++B-C \rightarrow$ etc.).^{4,8}

Wall and Porter⁹ have made the only investigation of the effect of change in the location of the energy barrier. The barrier was the symmetrical one appropriate to the $H+H_2$ reaction. It was shifted to four locations along a line *perpendicular* to the reaction coordinate, i.e., retaining the H-H-H symmetry. Trajectories were run in one dimension ("1D"; collinear reaction). It was found that the barrier location made little difference to the effective activation energy, but had a marked effect on the efficiency with which initial translational energy was converted into product vibration. These generalizations must be applied with caution in predicting the 2D or 3D behavior; see Sec. IV.B below.

A variable that has not previously been examined under category (2) is the location of the energy barrier along the reaction coordinate. In the following paper¹⁰ evidence is presented which indicates that in related series of reactions, with increasing barrier height the

¹ F. T. Wall, L. A. Hiller, Jr., and J. Mazur, *J. Chem. Phys.* **29**, 255 (1958); **35**, 1284 (1961).

² N. C. Blais and D. L. Bunker, *J. Chem. Phys.* **37**, 2713 (1962). For later work see M. Karplus in *Structural Chemistry and Molecular Biology*, A. Rich and N. Davidson, Eds. (W. H. Freeman and Co., San Francisco, 1968); E. F. Greene and J. Ross, *Science* **159**, 587 (1968).

³ (a) M. Karplus, R. N. Porter, and R. D. Sharma, *J. Chem. Phys.* **40**, 2033 (1964); (b) **43**, 3259 (1965).

⁴ N. Blais, *J. Chem. Phys.* **49**, 9 (1968); M. Godfrey and M. Karplus, *ibid.* **49**, 3602 (1968).

⁵ J. C. Polanyi, *Transfert d'Énergie dans les Gaz*, R. Stoops, Ed. (Interscience Publishers, Inc., New York, 1962), pp. 177, 526.

⁶ P. J. Kuntz, E. M. Nemeth, J. C. Polanyi, S. D. Rosner, and C. E. Young, *J. Chem. Phys.* **44**, 1168 (1966).

⁷ For the effect of reagent mass on a general interaction potential see D. L. Bunker and N. C. Blais, *J. Chem. Phys.* **39**, 315 (1963); compare **41**, 2377 (1964).

⁸ (a) P. J. Kuntz, E. M. Nemeth and J. C. Polanyi, *J. Chem. Phys.* **50**, 4607 (1969); (b) P. J. Kuntz, M. H. Mok, and J. C. Polanyi, *J. Chem. Phys.* **50**, 4623 (1969).

⁹ F. T. Wall and R. N. Porter, *J. Chem. Phys.* **39**, 3112 (1963).

¹⁰ M. H. Mok and J. C. Polanyi, *J. Chem. Phys.* **51**, 1451 (1969), following paper; referred to as Paper II.

barrier location moves to successively later positions along the reaction coordinate. It is therefore important to know the effect on the reaction dynamics of a shift in barrier location along this coordinate. In the present paper this is discussed in relation to atomic exchange reactions $A+BC \rightarrow AB+C$, where the masses of the particles are $m_A=m_B=m_C$. In a subsequent paper the effect of changing mass combination on the generalizations for the equal-mass case, and the effect of initial rotation in the molecular reagent, will be examined.^{11a} The effect of shifting barrier location is also being explored for the case of four-atom exchange reaction $AB+CD \rightarrow AC+BD$.^{11b}

II. POTENTIAL-ENERGY FUNCTION

The potential-energy hypersurface was the extended LEPS equation described in Ref. 6,

$$E(r_1, r_2, r_3) = \frac{Q_1}{1+a} + \frac{Q_2}{1+b} + \frac{Q_3}{1+c} - \left(\frac{J_1^2}{(1+a)^2} + \frac{J_2^2}{(1+b)^2} + \frac{J_3^2}{(1+c)^2} - \frac{J_1 J_2}{(1+a)(1+b)} - \frac{J_2 J_3}{(1+b)(1+c)} - \frac{J_1 J_3}{(1+a)(1+c)} \right)^{1/2}. \quad (1)$$

The coordinates r_1 , r_2 , and r_3 refer to the AB, BC, and AC internuclear separations. The Coulombic and exchange integrals for each pair of atoms, Q and J , were obtained from the Morse function for that pair of atoms and from the repulsive analogue of the Morse function. It was necessary, therefore, to ascribe an equilibrium separation r^0 , a Morse constant β ,¹² and a bond dissociation energy D_e (abbreviated here to D) to each pair of atoms. The constants for H_2 were used in this work for two of the three pairs of atoms, namely for AB and BC. For AC, r and β (i.e., r_3 and β_3) were again those for H_2 , but D (D_3) was decreased from 109 to 79 kcal mole⁻¹. This was done in order to render the reaction $A+BC \rightarrow AC+B$ endothermic by 30 kcal mole⁻¹, and hence to ensure that in the energy range used in this work the reaction under study was $A+BC \rightarrow AB+C$, exclusively.

The constants a , b , and c were chosen so as to obtain an asymmetric location for the crest of energy barrier. It was found that for $a=c < b$ the barrier was along the "approach coordinate" r_1 , and for $b=c < a$ the barrier was along the "retreat coordinate" r_2 . In order to switch the barrier in a symmetrical fashion from the approach to the retreat coordinate, the magnitudes of b and a were exchanged. The constants of Eq. (1) for surface I were: $r_1^0=r_2^0=r_3^0=0.742$ Å, $\beta_1=\beta_2=\beta_3=1.942$

Å⁻¹, $D_1=D_2=109.435$ kcal mole⁻¹; $D_3=79.435$ kcal mole⁻¹, $a=c=0.05$, $b=0.30$. The constants for surface II were the same as for surface I but with $b=c=0.05$, $a=0.30$. Only these two surfaces were used. On surface I the crest of the barrier was located at $r_1^\ddagger=1.20$ Å, $r_2^\ddagger=0.80$ Å; on surface II it was at $r_1^\ddagger=0.80$ Å, $r_2^\ddagger=1.20$ Å.¹³

Though the computations permit the particles to move freely in three dimensions, the LEPS function in general favors collinear approach. Where the barrier is in the retreat coordinate the LEPS function will favor collinear retreat. Special interest, therefore, attaches to the potential-energy surface for collinear reaction.

The collinear surface is shown in Fig. 1(a) for surface I, and in Fig. 1(c) for surface II. The classical barrier height is seen to be $E_c=7.0$ kcal in either case, but for surface I the barrier is in the approach coordinate and for surface II the barrier is in the retreat coordinate.¹⁴ In Figs. 1(b) and 1(d) the potential-energy changes for collinear approach of A to BC (with BC held at equilibrium separation, $r_2=r_2^0$) can be read off the abscissa. It is evident that on surface I the repulsion at r_1^0 is very significant whereas on surface II it is negligible. This reflects the fact that on surface I the activation barrier is met as $r_1 \rightarrow r_1^0$, whereas on surface II the barrier is not encountered till $r_1 \approx r_1^0$ and $r_2 > r_2^0$. The enhanced repulsion at r_1^0 on surface I compared with II can be seen to be even more marked when the approach of A is no longer collinear with BC. The characteristic difference between surfaces I and II is not, therefore, restricted to collinear approach.

III. COMPUTATION PROCEDURE

The classical Hamiltonian equations of motion were integrated numerically, for motion in 3D. The method was the same as that used in earlier work.^{8b} The equations of motion were reduced to 12 simultaneous differential equations by eliminating only the motion of the center of mass. The Adams-Moulton method was used for integration. The value of the time interval was $\delta t=3 \times 10^{-16}$ sec. The trajectories were required to have energy conserved to better than 1 in 10^8 . Checks were made to ensure that for sample trajectories product energy and angular distribution were substantially invariant with decreasing δt . [An example is cited in the caption to Fig. 3(b)].

¹³ Since the barrier is located asymmetrically ($r_1 \neq r_2$), neither surface I nor II is appropriate to the exchange reaction $H+H_2$. The spectroscopic constants for H_2 were used simply in order to obtain qualitatively reasonable ranges of interaction on these hypothetical energy surfaces.

¹⁴ It may not be clear why it was thought worthwhile to investigate surface II separately from surface I, in view of the fact that trajectories on surface II must be the reverse of those on I. The explanation lies in the fact that the product energy distributions and angular distributions from the surface I trajectories did not correspond to the reagent distributions of greatest interest for surface II. Consequently, additional trajectories were required.

¹¹ (a) B. A. Hodgson, J. C. Polanyi, and W. H. Wong, J. Chem. Phys. (to be published); (b) M. H. Mok and J. C. Polanyi, J. Chem. Phys. (to be published).

¹² G. Herzberg, *Spectra of Diatomic Molecules* (D. Van Nostrand Co., Inc., Princeton, N.J., 1950), p. 532.

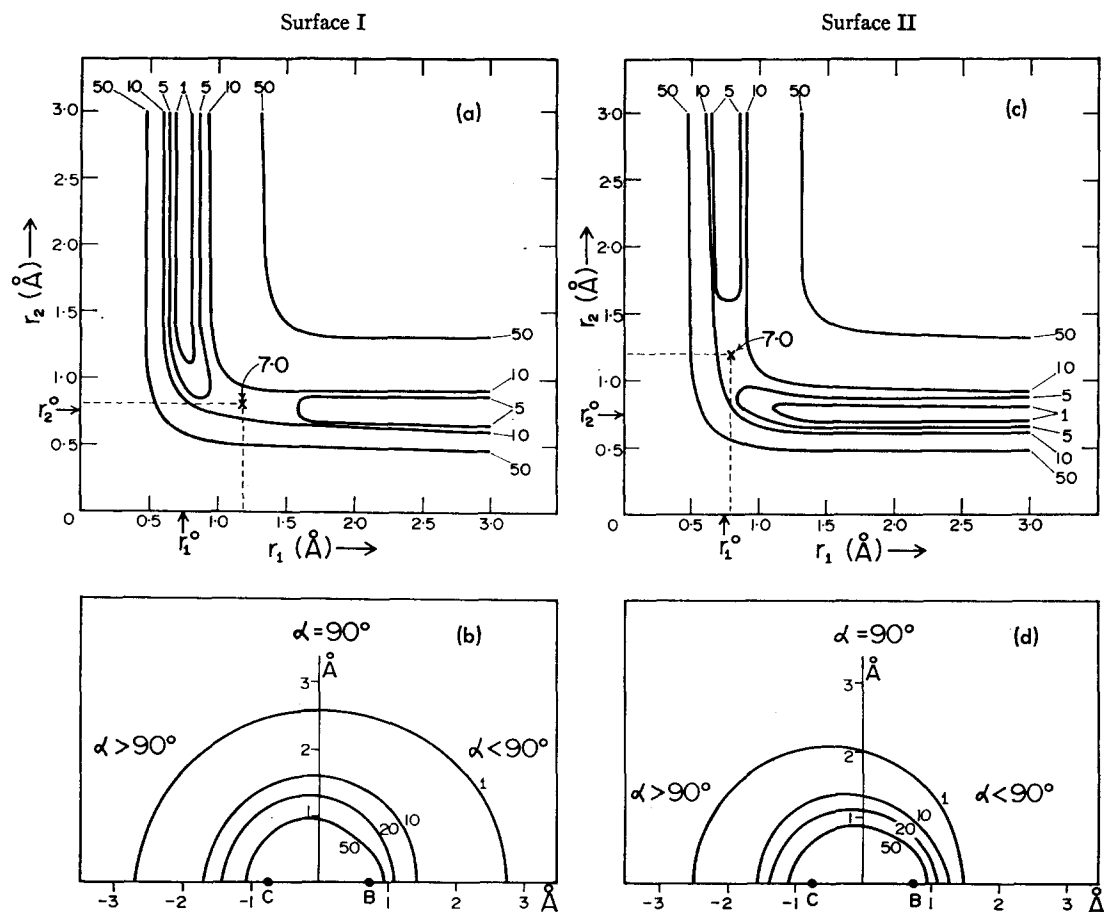


FIG. 1. All energy contours are in kilocalories mole⁻¹ (a) Potential-energy surface I, for collinear reaction. (b) Potential-energy contours on surface I for the approach of A from various angles α , with the BC separation held at its equilibrium value r_2^0 . (For collinear approach, $\alpha = 0^\circ$.) (c) Potential-energy surface II, for collinear reaction. (d) Potential-energy contours on surface II for the approach of A from various angles, with BC at r_2^0 .

The initial conditions were varied systematically. The spatial variables describing the initial conditions for a given trajectory were those described previously.^{8b} The molecule under attack, BC, lay in the xy plane with its center of mass at the origin, and made an angle α with the x axis. The initial BC separation was always taken as the equilibrium separation r_2^0 . Atom A lay on the x axis at a distance R from the origin. Initially $R = 6$ Å. Reaction was assumed to be complete when the distance separating the departing atom C from the center of mass of AB exceeded 6 Å. The velocity vector of A was specified in magnitude by the initial relative translational energy of the reactants E_{trans}^i , and in direction by the impact parameter b and the azimuthal angle δ . In selecting trajectories, those of large α were weighted according to $\sin \alpha$, trajectories of large b according to b . When initial vibration E_{vib}^i was introduced into BC, the phase of vibration was either such as to cause BC to expand from r_2^0 (termed "positive") or contract from r_2^0 . The effect of initial rotation in BC was not examined. Only single values, rather than distribution functions, were used for E_{trans}^i

and E_{vib}^i in calculating reaction cross sections and histograms.

The computations reported here were restricted to the single mass combination $m_A = m_B = m_C = 1$ amu. It should be noted, however, that the results are equally valid for any other mass, provided $m_A = m_B = m_C$.⁶ A trajectory generally took 400 to 700 iterations, corresponding to an interaction time of $1\text{--}2 \times 10^{-13}$ sec. The computer time per trajectory was ~ 10 sec. A "batch" of trajectories, corresponding to a particular set of initial conditions, comprised ~ 300 trajectories (where two phases of vibration had to be included, a "batch" was double this size).

Two sets of data, corresponding to two methods of choosing the initial conditions, were computed on each energy surface. These two types of initial conditions will be referred to as *nonequilibrium* and *equilibrium*.

The nonequilibrium initial conditions on surface I consisted in the assignment of the total reagent energy to relative translation. The nonequilibrium initial conditions for surface II assigned all but 1.5 kcal mole⁻¹ of the total reagent energy to vibration in the molecule

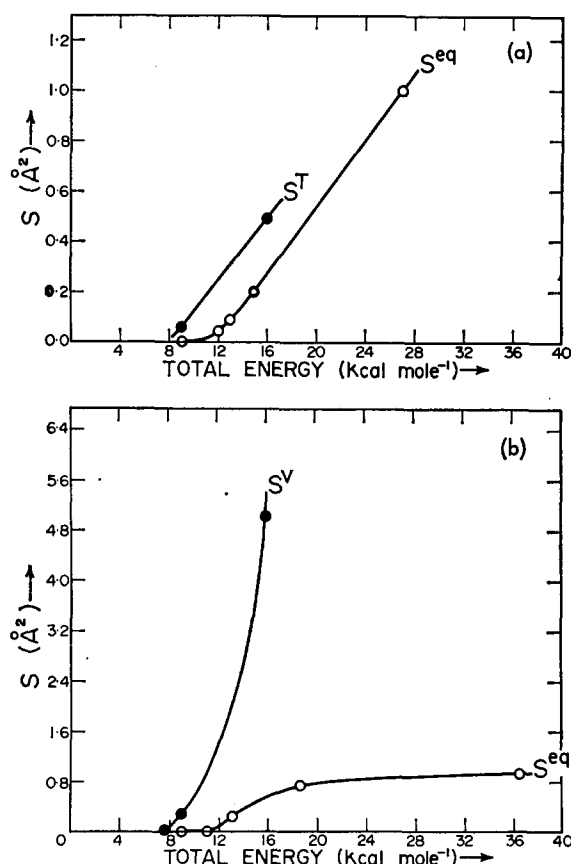


FIG. 2. (a) Reactive cross sections on surface I. The curve labeled S^T gives the cross section as a function of E_{trans}^i . The curve labeled S^{eq} gives the cross section as a function of $E_{\text{trans}}^i + E_{\text{vib}}^i$. (b) Reactive cross sections on surface II. The curve labeled S^V gives the cross section as a function of E_{vib}^i plus a constant 1.5 kcal mole $^{-1}$ of E_{trans}^i . The curve labeled S^{eq} gives the cross section as a function of $E_{\text{trans}}^i + E_{\text{vib}}^i$.

under attack, the 1.5-kcal mole $^{-1}$ balance of the initial energy being assigned to relative translation (since the reagents must approach one another).

The equilibrium initial conditions were defined in an identical fashion for surfaces I and II. The energies of translation and vibration were regarded as being the means of Boltzmann distributions, and were made to correspond to the same temperature T . When a value

was assigned to E_{trans}^i a temperature was calculated from $T = (2/3R) E_{\text{trans}}^i$. The vibrational energy was then calculated from¹⁵

$$E_{\text{vib}}^i = [Nh\nu / (e^{h\nu/kT} - 1)] + \frac{1}{2}(Nh\nu), \quad (2)$$

where ν is the frequency of oscillation in the simple harmonic approximation. In applying Eq. (2) we arbitrarily took $h\nu/k = 1000^\circ\text{K}$, corresponding to a vibrational spacing of ~ 2 kcal mole $^{-1}$ (a rough median figure for a wide range of diatomic molecules).

IV. RESULTS AND DISCUSSION

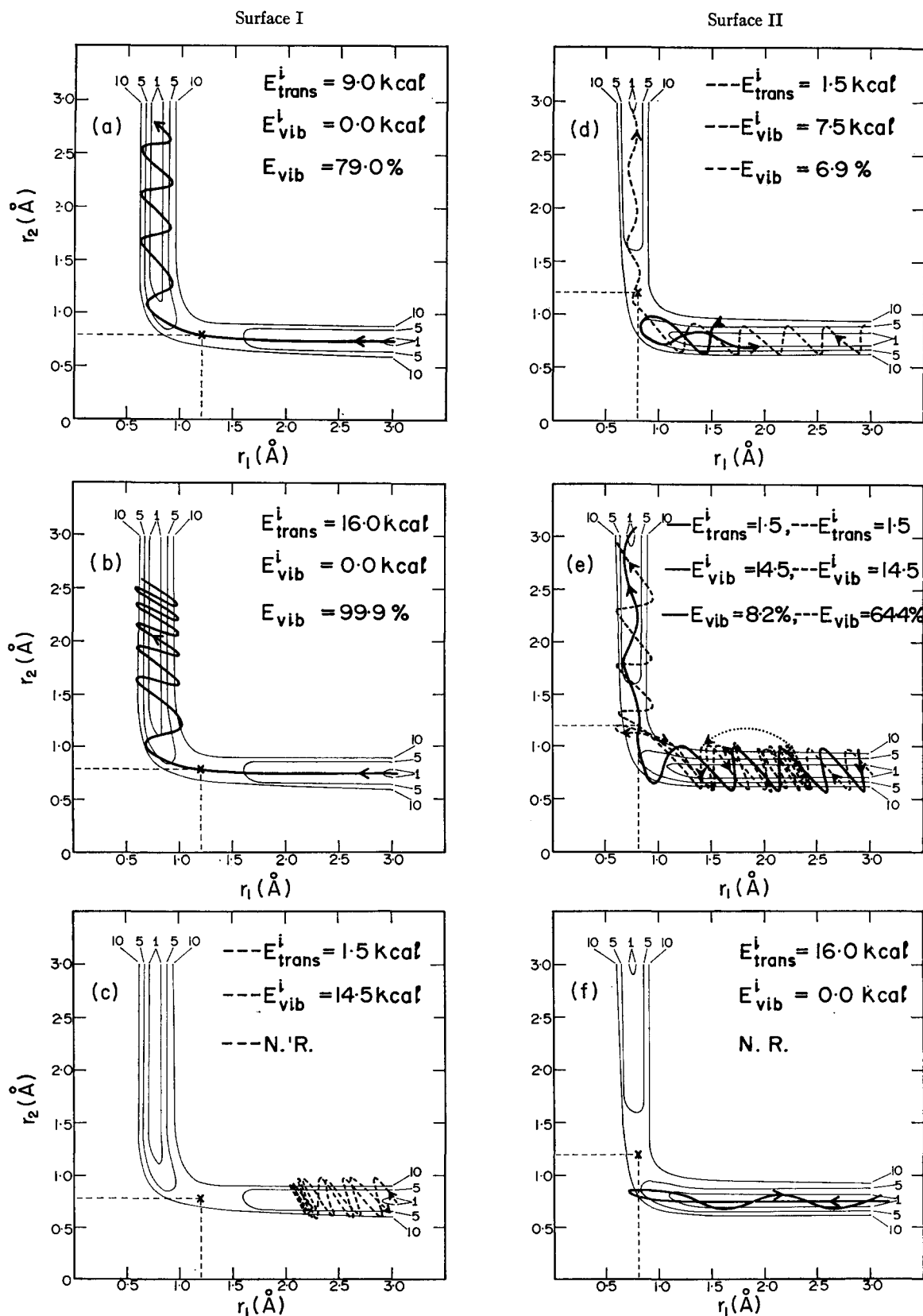
A total of 20 batches of trajectories were run. Each "batch" constituted a statistically meaningful group of trajectories, yielding a total reaction cross section, and also partial cross sections corresponding to distributions over product vibrational, rotational, and translational energies, and over product angles. The batches were equally divided between surface I and surface II. They were further equally subdivided, on each surface, into five batches using a nonequilibrium initial energy distribution, and five batches using an equilibrium initial distribution. For every energy used on surface II a complete computation was made with BC initially expanding, and then the calculation was repeated with BC contracting; finally the reactive cross sections were averaged, and all the trajectories, designated a single batch, were included in the histograms which gave product energy and angular distributions. The reactive cross sections calculated for the two opposite phases of vibration of BC generally differed by a factor of $\lesssim 2$; the average figure was therefore entirely adequate for the present purposes.

A. Reactive Cross Section

The reactive cross section as a function of total reagent energy, $S(E_{\text{tot}})$, is presented graphically in Fig. 2(a) for surface I, and in Fig. 2(b) for surface II. It is apparent for both surfaces that at any reagent energy a nonequilibrium distribution (S^T or S^V , upper curves) gives substantially greater probability of reaction than does an equilibrium distribution (S^{eq} , lower curves). The nature of the nonequilibrium distribution

FIG. 3. Collinear trajectories for nonequilibrium reagent energies. The reagents have traversed ~ 3 Å of their trajectory before the plots shown here begin. Where the initial phase of vibration is positive the trajectory is indicated by the solid line, where negative by a broken line. (a) Surface I; reagents have exclusively E_{trans}^i close to the threshold energy for reaction. (b) Surface I; reagents have exclusively E_{trans}^i , substantially in excess of threshold energy. The periodic reduction in r_2 as the reagents separate corresponds to atom B reaching out toward the departing C. (c) Surface I; reagents have almost exclusively E_{vib}^i . The positive initial phase of vibration gave a trajectory similar in appearance and outcome. (d) Surface II; reagents have almost exclusively E_{vib}^i , close to the threshold for reaction. The trajectory with positive vibrational phase and identical initial conditions failed to react. (In order to simplify the figure only a portion of this trajectory is shown.) (e) Surface II; reagents have almost exclusively E_{vib}^i , substantially in excess of threshold energy. The trajectory with negative initial phase resulted in approach of the reagents to $r_1 = 0.6$ Å, followed by separation to $r_1 = 2.4$ Å, and finally a second approach (not shown in detail, but implied by the curved dotted arrow) leading to reaction. (f) Surface II; reagents have exclusively E_{trans}^i . All these energy surfaces should have skewed axes in order that the motion of a rolling ball shall represent an allowed trajectory; the angle between the axes would then be 60° for the case of three equal masses. The trajectories were, however, obtained by integration of the equations of motion, using initial $\alpha = 0.0^\circ$, $\delta = 0.0^\circ$, and $b = 0.01$ Å. All these trajectories were checked with reduced δt , with more-detailed sampling, and with b varied in the range 0.001–0.1 Å.

¹⁵ R. C. Tolman, *The Principles of Statistical Mechanics* (Oxford University Press, London, 1938), p. 379.



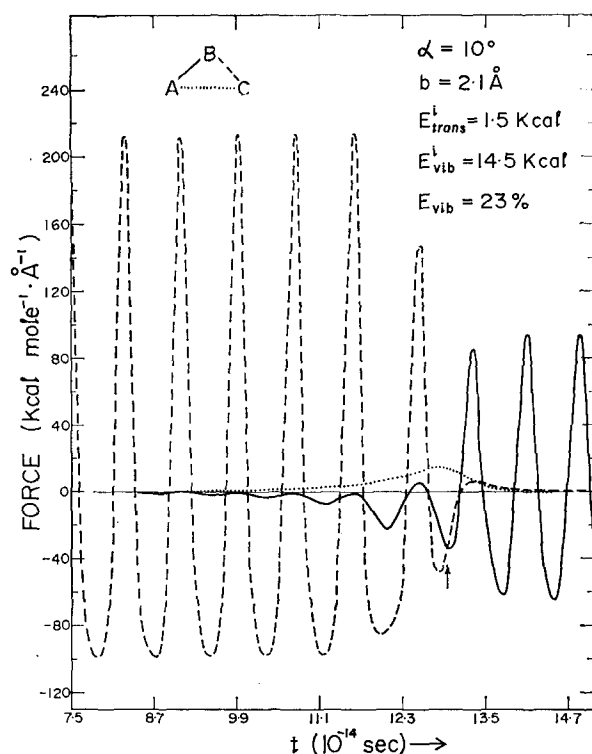


FIG. 4. Interatomic forces for a single trajectory at large impact parameter on surface II, as a function of time lapse. Solid lines indicate F_1 , dashed lines F_2 , and dotted lines F_3 . The molecule under attack, BC, was initially contracting ("negative phase"). The values of α and b cited in the figure are initial values. The initial value of δ was 0° .

that favors reaction is, however, quite different in the two cases; on surface I translation favors reaction, whereas on surface II vibration favors reaction.

Apart from the 15 batches of trajectories recorded in Figs. 2(a) and 2(b), five further batches were run. Three of these were on surface I. All three had $E_{\text{trans}}^i = 1.5$ kcal mole $^{-1}$, but increasing amounts of vibrational energy, namely $E_{\text{vib}}^i = 6, 7.5$, and 14.5 kcal mole $^{-1}$. The cross section was 0.0 to within ~ 0.0045 Å 2 , even for the highest vibrational energy. (The limits of error have been taken from the effect that would be produced by a change of 1 in the number of reactive trajectories. The number of reactive trajectories was zero in the present case.) Whereas the data of Fig. 2(a) illustrates the utility of translational energy in promoting reaction on surface I, this additional data illustrates the inutility of vibration. The evidence is striking, since a vibrational energy in excess of two times the barrier height produced no reactive trajectories whatever. The two other batches of trajectories (making up the total of 20) were used to investigate the effectiveness of translation in promoting reaction on surface II. The vibrational energy was held at $E_{\text{vib}}^i = 0$, and the translation was increased to $E_{\text{trans}}^i = 9$ and 16 kcal mole $^{-1}$. Despite the fact that in the latter case the translational energy exceeded two times the barrier

height, no reactive trajectories were observed; for both these initial conditions, therefore, $S = 0.0$ to within ~ 0.0045 Å 2 .

These findings are exemplified for the case of collinear reaction in Fig. 3. The trajectories are records of portions of actual collinear encounters, from $r_1 = 3$ Å to $r_2 = 3$ Å. The course of the trajectory can be understood by thinking of the path that a rolling marble would take across the surface.¹⁶ (For this to be a proper analogy it would be necessary for the axis of the energy surfaces to be skewed at 60° , for the case of three equal masses.) In Fig. 3(a) one sees that the momentum of a rolling ball travelling *parallel* to the entry valley of the energy surface tends to carry it over a barrier placed in that valley. This illustrates the effectiveness of E_{trans}^i on *surface I*. The ineffectiveness of E_{vib}^i on the same surface is evident in Fig. 3(c). Though this is, once again, an actual trajectory we shall account for it in terms of a rolling ball. The ball starts with largely *transverse* momentum, and attempts to cross the barrier at a place remote from the lowest crossing point. It has insufficient parallel momentum (this term is reserved for translational motion along the reaction coordinate) to carry it over the barrier at this unfavorable crossing point, and it is reflected.

The effectiveness of E_{vib}^i on *surface II* is evident from Fig. 3(d). When the vibrational phase is such as to permit the system to round the bend on the energy surface, then it can be seen that transverse momentum in the entry valley is converted to parallel momentum along the exit valley. This parallel momen-

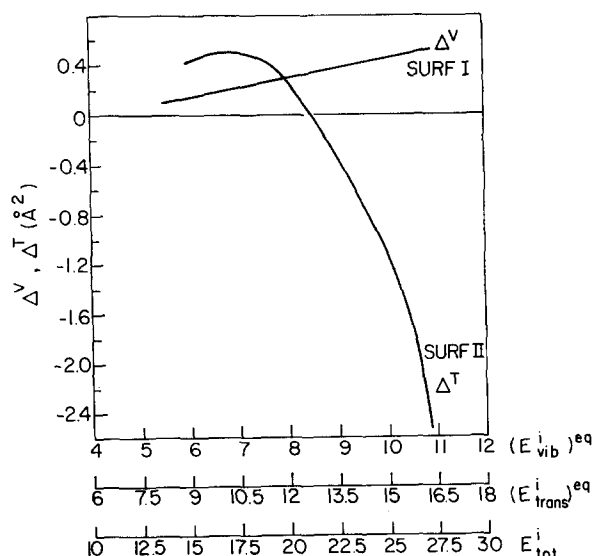


FIG. 5. The magnitudes of Δ^v and Δ^T in Eqs. (4) and (5) versus the equilibrium proportions of translational and vibrational energy in the reagents. The curves begin at the energies for which S^v and S^T are above threshold.

¹⁶ S. Glasstone, K. J. Laidler, and H. Eyring, *The Theory of Rate Processes* (McGraw-Hill Book Co., New York, 1941), pp. 10, 109.

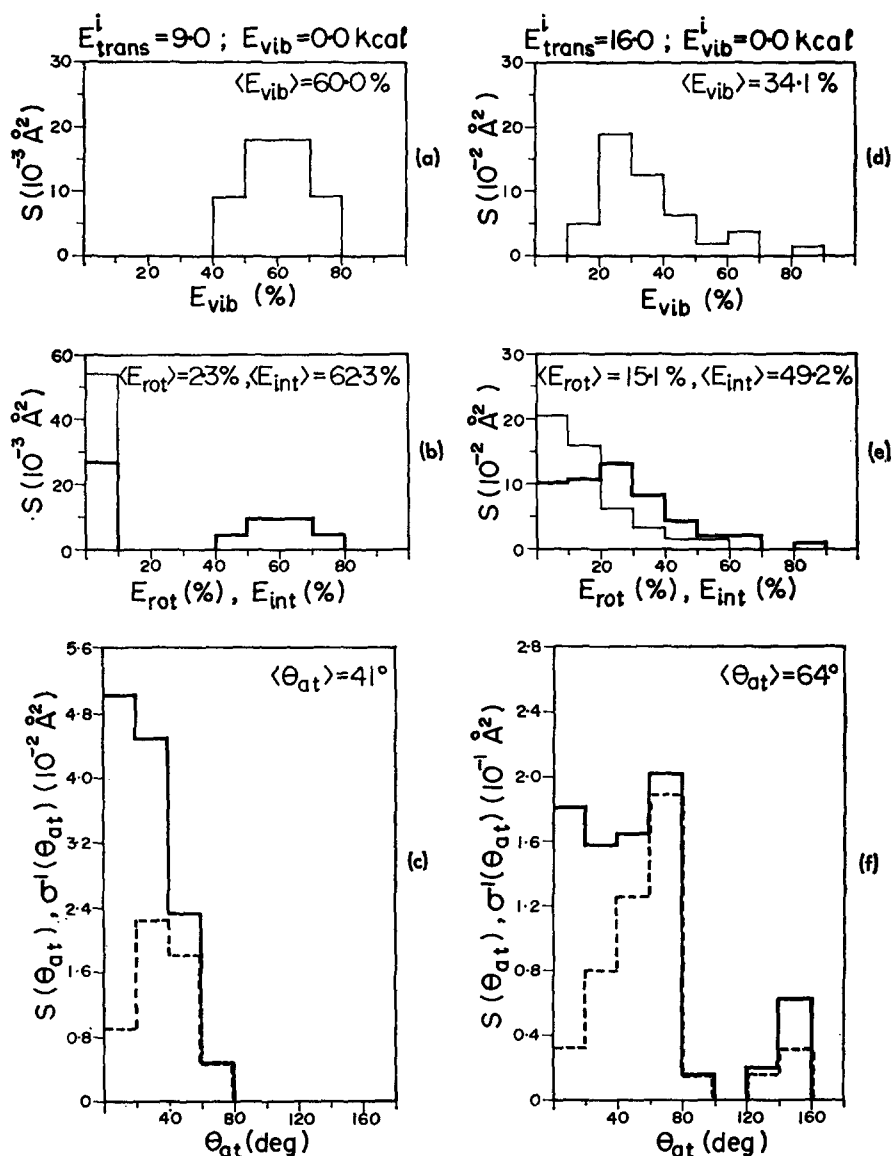


FIG. 6. Histograms of product energy distribution and angular distribution on surface I at low E^i_{trans} (left-hand column) and on surface I at high E^i_{trans} (right-hand column). The heavy line in the energy histograms (b) and (e) records E_{int} . In the angular histograms (c) and (f) the broken line indicates $S(\theta_{at})$, and the solid line $\sigma'(\theta_{at})$ [the latter quantity is 2π times the differential cross section $\sigma(\theta_{at})$, defined in Eq. (6)]. Forward scattering of the atomic product, and hence backward scattering of the molecular product, is at small θ_{at} .

tum is able to carry the rolling ball over the barrier located, on surface II, in the exit valley. The converse is true for E_{trans} on surface II: In Fig. 3(f) it is evident that parallel momentum along the entry valley becomes transverse momentum in the exit valley, and consequently is not able to carry the system over a barrier in that valley.

These collinear trajectories¹⁷ have illustrative value once the 3D behavior is known, but (see Sec. IV.B, below) it would be risky to use them as an exclusive guide to the over-all behavior.

The magnitude of the effects described above, for a displacement of the barrier of about 0.3 Å from the position expected for a symmetrical barrier,¹⁷ can be judged from the ratio of S^T/S^{eq} on surface I, and S^V/S^{eq} on

surface II, in the region of the threshold energy for reaction in the nonequilibrium system. The ratio was ~ 10 on surfaces I and II. Since in energy-restricted systems the majority of reactive collisions occur in the threshold region,^{3b} it is of particular interest to note the gain in rate to be expected in this region from a redistribution of the available energy. On surface I beyond threshold the gain from using a nonequilibrium distribution becomes fairly small (~ 2 times, here), and remains fairly constant. On surface II, however, at energies approximately double the threshold energy, the ratio $S^V/S^{eq} \sim 10$. The advantage of the nonequilibrium distribution becomes even larger at higher energies on surface II, increasing without a discernible limit.

It is noteworthy, also, that despite identical barrier heights on surfaces I and II, the absolute value of the

¹⁷ I. Shavitt, R. M. Stevens, F. L. Minn, and M. Karplus, J. Chem. Phys. **48**, 2700 (1968).

nonequilibrium cross section S^V for surface II is far greater at high energies than is S^T on surface I [note the fivefold reduction in scale of the ordinate in Fig. 2(b) as compared with 2(a)]. The explanation is to be found in the fact that when the B-C bond is extended (in the course of BC vibration) A and B start to attract from a considerable separation. This is why in the final stage of the complicated trajectory in Fig. 3(e) as BC vibrates the A atom is slowly pulled back towards B, i.e., the oscillating trajectory moves toward smaller r_1 .¹⁸ That this is a real phenomenon, rather than a weakness in the LEPS function, is indicated by the fact that in the limit of extremely high vibrational energy in BC, the B-C bond at full amplitude will be close to dissociated; in this case the attacking atom will be attracted to B over almost as large a range as if only atoms A and B were present.

The effect described in the previous paragraph has more relevance to increased reactive cross section when it occurs at large impact parameters (rather than at $b=0$, as in a collinear trajectory). In Fig. 4 we show a graph of the forces operating between the atoms during the course of a large-impact-parameter collision at high initial vibrational energy on surface II. The product, following this encounter, has high vibrational energy. To produce a "force plot"¹⁹ of this type we have arbitrarily separated the forces operative in the course of an encounter into the three components,

$$-(\partial E/\partial r_i)_{jk} = F_{ij}, \quad (3)$$

where i, j , and k can take on the values 1, 2, 3; $i \neq j \neq k$. The forces $F_1 \equiv F_{AB}$, $F_2 \equiv F_{BC}$, and $F_3 \equiv F_{AC}$, as well as the internuclear separations, are recorded as a function of time. Force plots on surface I (not given) differed from those on surface II in that they showed an increase in F_1 as reaction proceeded, i.e., AB repulsion, prior to $r_1 \approx r_1^\ddagger$. This is clearly linked to the presence of a barrier along the r_1 coordinate. There was no rise in F_1 on surface II, only an increase in F_3 at close approach. (The oscillation in F_2 on surface II is simply due to the BC vibration in the initial state.)

Figure 4 illustrates for a trajectory at large impact parameter that A-B becomes attractive as B-C is expanding. In the example given A-B starts to attract significantly at a separation of $\sim 2.2 \text{ \AA}$ (8.8×10^{-14} sec on the abscissa). Each time B-C expands the A-B

attraction increases. In this trajectory, at the symmetrical configuration $r_1 = 0.92 \text{ \AA}$, $r_2 = 1.05 \text{ \AA}$, the A-B and B-C attractive forces are the same (see arrow at 13.0×10^{-14} sec). Thereafter the B-C force diminishes asymptotically to zero as $r_2 \rightarrow \infty$. AB, having been formed as consequence of the long-range attraction between A and B, has significant energy in the A-B coordinate, i.e., it is vibrationally excited.

In order to obtain some measure of the contribution of the less favorable degree of freedom (which we have shown to be ineffective in isolation) under equilibrium conditions, we have calculated the increments in cross section Δ^V on surface I,

$$\Delta^V = S^{\text{eq}}(E_{\text{tot}}^i) - S^T(E_{\text{trans}}^i)^{\text{eq}}, \quad (4)$$

and Δ^T on surface II,

$$\Delta^T = S^{\text{eq}}(E_{\text{tot}}^i) - S^V(E_{\text{vib}}^i)^{\text{eq}}. \quad (5)$$

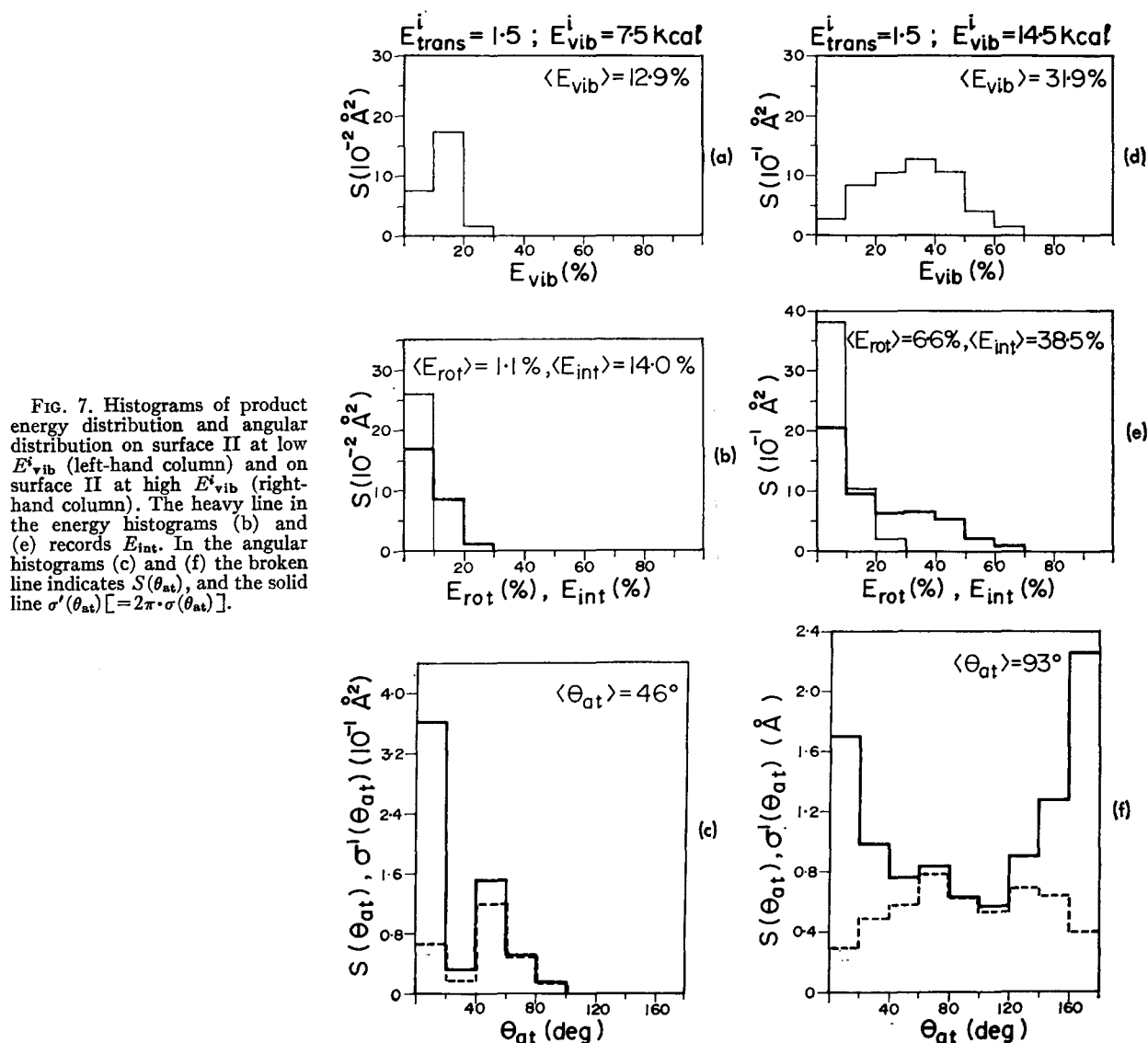
The first term on the rhs of (4) and of (5) is taken from the S^{eq} curves in Figs. 2(a) and 2(b), respectively. The second term on the rhs of (4) and (5) is taken from the S^T and S^V curves of Figs. 2(a) and 2(b).²⁰ The energy used in calculating the second term in (4) is the equilibrium fraction of E_{tot} that is present as translation on surface I, and in (5) the equilibrium fraction that is present as vibration on surface II.²⁰ Implicit in the manner of calculating Δ^V and Δ^T is the assumption that the favored degree of freedom can be regarded as being as effective in the presence of the equilibrium proportion of the disfavored degree of freedom, as it is when it is there in isolation. If we make this assumption then we are led (Fig. 5) to the following conclusions: (1) Under equilibrium conditions on surface I the disfavored degree of freedom E_{vib}^i makes a significant contribution to the total reactive cross section. This contribution increases steadily with increasing total energy; (2) under equilibrium conditions on surface II the disfavored degree of freedom E_{trans}^i increases the total reactive cross section in the region of total energy up to roughly two times the threshold energy of $S^{\text{eq}}(E_{\text{tot}}^i)$. At higher E_{tot}^i than this, the disfavored degree of freedom significantly and increasingly *diminishes* the total reactive cross section, i.e., $\Delta^T < 0$.

It should be stressed that Fig. 3 is to be used for comparison of reactive cross sections at a *given total energy*: (i) with all of that energy in the favored mode of motion for reaction, as compared (ii) with that

¹⁸ The important category of trajectories represented by the broken line in Fig. 3(e) might easily be overlooked if one attempted to visualize trajectories in terms of a rolling ball (the more so if the axes were skewed). The reason that the rolling ball could trace out such a path is probably that the entry valley is broader at smaller r_1 than at large r_1 . Consequently, when the reflected ball is moving with small momentum toward increasing r_1 (and reaction seems about to fail) it can be reflected back once again toward the corner of the potential-energy surface.

¹⁹ P. J. Kuntz, E. M. Nemeth, J. C. Polanyi, and W. H. Wong, "Distribution of Reaction Products (Theory). VI. Hot-Atom Reactions, T+HR," J. Chem. Phys. (to be published). This paper deals with the dynamics of hot-atom reactions, and includes a discussion of force plots.

²⁰ For simplicity we treat $S^V(E_{\text{vib}}^i)$, recorded in Fig. 2(b), as if it were the cross section for reaction with only vibrational energy present in the reagents. In fact, in order to bring the reagents within range of one another, $1.5 \text{ kcal mole}^{-1}$ of the energy loosely designated E_{vib}^i was invariably present as initial translation. In computing Δ^T we have treated the ordinate of Fig. 2(b) as if it recorded vibrational energy only. The total translational energy which contributes to $S^{\text{eq}}(E_{\text{tot}}^i)$ exceeds the translational energy hidden in E_{vib}^i by a factor of 5-15 times; consequently, Δ^T does throw light on the role of translation under equilibrium conditions.



same energy divided in equilibrium proportions between translation and vibration. Figure 5 is to be used for comparison of reaction cross sections at different total energies: (i') with a given energy in the favored mode of motion, compared (ii') with that same energy in the favored mode of motion plus an additional amount of energy, corresponding to the equilibrium proportion, in the disfavored mode of motion. The increase in reactive cross section in going from (i') to (ii') is given in Fig. 5, where it is expressed as a function of the amount of energy in the favored mode in (i'), and, at the same time, as a function of the additional amount of energy in (ii').

As already noted, only the effects of changing the translational and vibrational energy (separately and concurrently) have been considered in this study. It seems unlikely that the major qualitative conclusions regarding the superiority of E_{trans}^i on surface I and of

E_{vib}^i on surface II (under nonequilibrium conditions) would be altered by the inclusion of rotation E_{rot}^i as a third alternative in the comparison. The contrast in behavior between S^{eq} on surface I and II might be less marked if E_{tot}^i included in each case not only $E_{trans}^i + E_{vib}^i$, but also E_{rot}^i . But here again the qualitative difference would be expected to survive in view of the marked superiority of E_{trans}^i to E_{vib}^i on surface I, and the converse effect on surface II.²¹

B. Product Energy Distribution

Figure 6 gives histograms of product vibrational, rotational, and total internal energy distributions for surface I. The histograms of product translational exci-

²¹ M. Karplus, R. N. Porter, and R. D. Sharma, Ref. 3(b), investigated the effect of increasing amounts of rotational energy in the molecule under attack, for the exchange reaction $H+H_2$; the effect of the rotational energy was slight.

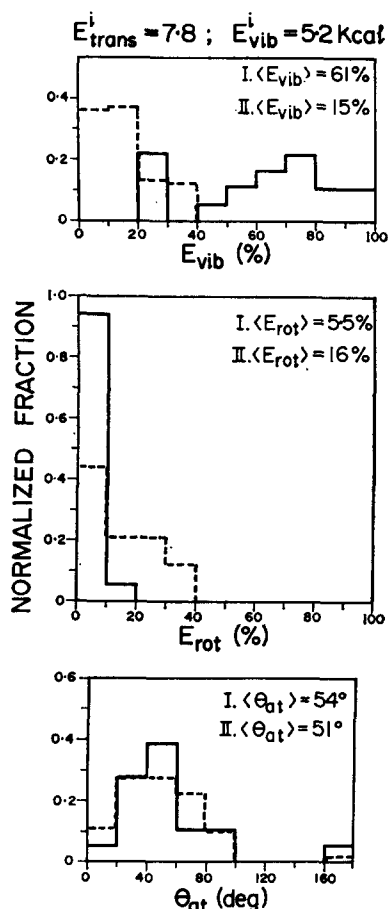


FIG. 8. Histograms of product energy distribution and angular distribution [$S(\theta_{at})$] with equilibrium proportions of initial translation and vibration; $E_{trans}^i = 7.8$ kcal mole $^{-1}$, $E_{vib}^i = 5.2$ kcal mole $^{-1}$. The broken line gives the results on surface I, the solid line on surface II.

tation (not shown) are simply the internal-energy histograms reflected in the 50% axis. The product energy distributions are expressed as absolute cross sections for the formation of product with a stated percentage of the total initial energy in a specified mode of motion in the product. To be consistent with earlier notation these partial cross sections should be symbolized $S^T(E_{vib} | E_{trans}^i)$, $S^T(E_{rot} | E_{trans}^i)$, and $S^T(E_{int} | E_{trans}^i)$. The histograms sort percent (E_{vib}/E_{tot}), percent (E_{rot}/E_{tot}), and percent (E_{int}/E_{tot}) into bins which accommodate intervals of a breadth equal to 10%. The initial translational energy is exemplified by two values only, in Fig. 6: The left-hand column [(a)–(c)] is for E_{trans}^i close to threshold, and the right-hand column [(d)–(f)] is for E_{trans}^i of approximately two times threshold.

The observation concerning product energy distribution on surface I can be summarized as follows. At a reagent energy close to threshold a major portion of E_{trans}^i (mean of $\sim 60\%$) is channeled into product vibration. The balance is largely in product translation. At increased reagent energy a *smaller* proportion of E_{trans}^i is channeled into product vibration, with a

commensurate increase in both product rotation and translation.

Inspection of the collinear trajectories on surface I at low and high E_{trans}^i [Figs. 3(a) and 3(b)] shows that the percentage conversion to vibration has *increased* with increasing collision energy, instead of decreasing. Since the collinear trajectories are even less representative at high collision energy than at low (due to the increased cone of reaction), this atypical behavior exhibited by the collinear trajectory is not too surprising. The increase in product rotation with collision energy for the full batch of trajectories on surface I is likely to be connected with the increase in initial orbital angular momentum (which is zero for collinear collisions at any energy). The increase in product translation may have to do with a tendency for the representational point in nonlinear collisions to cross the energy barrier elsewhere than at the lowest point. The rolling ball can then be deflected in the course of crossing the barrier in the entrance valley. The resultant momentum perpendicular to the entrance valley (of the energy surface for bent approach) becomes parallel momentum along the exit valley; that is to say, product translation.

On surface II the product energy distribution, and the effect of increases reagent energy on the product energy distribution, is similar to that on surface I with *the roles of translation and vibration interchanged at every point* (see Fig. 7). This *complementarity* has its origins in the fact that the favored mode of motion in the reagents changes from translation, on surface I, to vibration on surface II. The replacement of parallel motion (surface I) by perpendicular motion (surface II) in the reagents results, after the system rounds the bend in the surface, in the replacement of perpendicular (vibrational) for parallel (translational) motion in the products.

On surface I at a reagent energy just above threshold the product has substantial vibrational excitation and little translation; on surface II it has substantial translational excitation and little vibration. The detailed explanation is suggested by Fig. 3(d). The initial energy on surface II being vibration in BC this relative motion of B with respect to C becomes translation once the BC bond dissolves (i.e., when the system "rounds the bend" on the energy surface). The relative motion of B with respect to C is only sufficient to carry the system over the barrier in the retreat coordinate if the trajectory crosses the barrier at its lowest point. At this lowest point there is no appreciable force perpendicular to the reaction coordinate; hence the rolling ball continues to move parallel to the retreat coordinate, and there is little vibration in the product.

With an initial energy on surface II almost double the threshold energy the product translation decreased, whereas vibration increased; this was the converse of the behavior on surface I. The explanation [Fig. 3(e)] is that the initial vibration produces sufficient transla-

tion in the retreat coordinate that the rolling ball can cross the barrier at other than its lowest point. As the ball crosses the barrier it is therefore subject to a force perpendicular to the reaction coordinate that causes it to oscillate along r_1 , i.e., there is vibrational excitation in the new bond.

In terms of the atoms rather than the surface these explanations can be restated as follows: in the high-energy trajectories on surface II AB tends to be formed in a state of extension or compression (since the barrier can be crossed with r_1 compressed or extended) and hence is left vibrationally excited. On surface I at high energy it is not AB that is formed in a state of extension or compression, but BC that is dissolved from a state of compression or extension, with consequent enhancement in translation.

In Fig. 8 a comparison is made between the product energy distribution on surfaces I and II using an identical *equilibrium* initial energy distribution on each surface. The total energy was 13 kcal mole⁻¹, just above threshold for S^{eq} . The vibrational distribution on each surface closely resembled that obtained under nonequilibrium conditions, though with some broadening. The rotational excitation was somewhat increased on both surfaces, as compared with the nonequilibrium reactions. The general picture is sufficiently similar to that for the nonequilibrium case to suggest that the *dominant* reaction paths resemble those that have been described above to account for S^{T} on surface I, and S^{V} on surface II. The detailed dynamics must be more complicated, as indicated by the significant magnitudes of Δ^{V} and Δ^{T} (on surfaces I and II, respectively), and as suggested by the broadened product energy distributions.

C. Product Angular Distribution

Figures 6(c) and (f) record the product angular distributions on surface I at low and at high reagent energy. Two partial cross sections are shown; $S(\theta_{\text{at}})$ and $\sigma(\theta_{\text{at}})$. Both refer to the angular distribution of the atomic product (C) in the center-of-mass (cm) frame. The angle θ_{at} is measured relative to the cm direction of approach of atom A. A large value of θ_{at} corresponds, therefore, to "backward" scattering of the atomic product and hence "forward" scattering of the molecular product; $\theta_{\text{mol}} = 180^\circ - \theta_{\text{at}}$. The quantity $S(\theta_{\text{at}})$ is the cross section for the formation of atomic product within each of the nine intervals of θ_{at} without regard to the value of the out-of-plane angle δ . The mean angle $\langle\theta_{\text{at}}\rangle$ has been calculated from the distribution of $S(\theta_{\text{at}})$. The differential cross section $\sigma(\theta_{\text{at}})$ is related to $S(\theta_{\text{at}})$ through²²

$$\sigma(\theta_{\text{at}}) = S(\theta_{\text{at}}) / 2\pi \sin\theta_{\text{at}}. \quad (6)$$

The differential cross section is more easily derived from the angular distributions measured in experiments with crossed molecular beams than is $S(\theta_{\text{at}})$.

The observations regarding $\sigma(\theta_{\text{at}})$ on surface I are the following: The atomic product is scattered in the forward hemisphere, the molecular product backwards. At increased reagent energy E_{trans} the distribution of the molecular product shifts forward. This is a common observation at higher collision energies, and arises from the fact that the products interact for a briefer period^{2,3,8b,23} (in the limit of very high collision energy the AB·C interaction as the products separate becomes unimportant, and AB is scattered forward by "stripping"²⁴).

On surface II at low reagent energy the form of $\sigma(\theta_{\text{at}})$ is similar to that on surface I. At higher reagent energy the distribution of the molecular product shifts forward, as before. The shift is, however, more marked than on surface I.

In the case of surface II there is no precedent for the shift to more forward scattering of the molecular product at higher reagent energy, since the reagent energy is not relative translation but internal energy of the molecule under attack. The explanation of the shift is likely to lie in a diminution in the average AB·C repulsion between the products.²³ At higher reagent energy BC is more highly vibrating. We have already noted (Sec. IV.A and Fig. 4) that atom A tends to make its close approach to BC as BC is expanding. If the mean extension of B-C at the time that A comes within close range is greater at higher reagent energy, then the extent of AB·C repulsion will be diminished.

Figure 8(c) compares product angular distributions [$S(\theta_{\text{at}})$ only] on surfaces I and II using an identical *equilibrium* initial energy distribution on each surface. The total energy was just above threshold for S^{eq} . The means of the distributions are very similar to those on the nonequilibrium surface, though there is some increase in the breadth. This suggests once again (cf. Sec. IV.B) that the dominant modes of reaction resemble those identified in the nonequilibrium trajectories, with contributions from more complicated encounters.

D. Absolute Reaction Rate Theory

The theory of absolute reaction rates¹⁶ calculates reaction rate from the partition function of the activated state compared with the initial state. Surfaces I and II used in the present work differ predominantly in the configuration of the activated state, rather than in the shape of the surface in the neighborhood of this state. The transposition in the configuration of the activated state from the approach to the retreat coordinate, in passing from surface I to surface II, should alter the activated complex from H···H-H to H-H···H in a symmetrical fashion, without any

²³ J. C. Polanyi, *Discussions Faraday Soc.* **44**, 293 (1967).

²⁴ A. Henglein and G. A. Muccini, *Z. Naturforsch.* **17a**, 452 (1962); **18a**, 753 (1963); A. Henglein, K. Lacmann, and B. Knoll, *J. Chem. Phys.* **43**, 1048 (1965).

²² M. Karplus and L. M. Raff, *J. Chem. Phys.* **41**, 1267 (1964).

major change in the partition functions. Nonetheless the form of $S^{\text{eq}}(E_{\text{tot}}^{\ddagger})$ is different in the two cases. Close to threshold the difference is slight. Well above threshold the difference is substantial, in that on surface I $S^{\text{eq}}(E_{\text{tot}}^{\ddagger})$ is still rising almost linearly with $(E_{\text{tot}}^{\ddagger})$ at an energy at which $S^{\text{eq}}(E_{\text{tot}}^{\ddagger})$ on surface II shows a negligible increase with $E_{\text{tot}}^{\ddagger}$. The energy at which the cross section on surface II levels out is approximately two times the threshold energy for S^{eq} . This marked difference in behavior relates to individual cross sections, and should be of interest in connection with detailed versions of absolute rate theory, in particular that developed by Marcus.²⁵

There has been considerable discussion²⁶ of "adiabaticity" as a sufficient criterion for the detailed applicability of absolute rate theory. Adiabaticity, in this context, is the requirement that motion along the reaction coordinate shall neither contribute to nor gain from motion perpendicular to the reaction coordinate. Since absolute rate theory is based on properties of the initial and the activated states, it is, however, enough if the adiabatic criterion is satisfied until the system reaches the crest of the barrier.²⁷ Marcus has discussed the failure of adiabaticity (increasing with increasing collision energy) for collinear reaction on surfaces with symmetrical energy barriers, and has proposed a form for the nonadiabatic correction.^{27d}

On *surface I* it is evident that initial *translation* $E_{\text{trans}}^{\ddagger}$, which is responsible for the motion along the reaction coordinate, is the degree of freedom that is by far the most effective in carrying the system over the barrier.²⁸ This initial translation has been shown to contribute to product vibration (especially near threshold collision energy; see Sec. IV.B above). However, since this conversion of initial motion along the reaction coordinate into motion perpendicular to it takes place beyond the barrier, it does not have any bearing on the applicability of the absolute rate theory to the forward reaction. Initial *vibration* $E_{\text{vib}}^{\ddagger}$ [not involved in Figs. 3(a) and 3(b), but present in all the S^{eq} trajectories], constitutes motion perpendicular to the reaction coordinate before and at the barrier. After the barrier it appears to be divided between product vibration and translation; however, its fate at this stage of the reaction is, once again, irrelevant in the present context.

On *surface II* it is clearly initial vibration that is mainly responsible for carrying the system across the

barrier. Since motion across the barrier is, by definition, motion along the reaction coordinate, we have a situation in which motion that was initially perpendicular to the reaction coordinate, i.e., initially *vibration*, is being converted to an important extent into motion along the reaction coordinate prior to the crest of the barrier. From the effect on the reaction cross section of including significant initial *translation* (see Sec. IV.A, above) we surmise that the fate of this motion is to contribute mainly to vibration and rotation at the crest of the barrier, rather than to translation across the barrier. This is to be expected since there is clear evidence on surface I that relative translation of atoms A and B contributes to the internal motion in AB.

We sum up as follows: On both surfaces I and II there is a tendency for motions to become transposed as the representational point rounds the bend on the energy surface: motion initially perpendicular to the reaction coordinate becomes motion parallel to it, and vice versa. Since this transposition takes place as the system rounds the bend, it has the consequence that the adiabaticity criterion (for absolute rate theory) is met by surfaces with their barriers in the approach coordinate, but not by surfaces with their barriers in the retreat coordinate.

If the transposition of motions is quantitative on surface II by the time that the barrier in the exit valley is reached, it should be possible to apply the detailed absolute rate theory in a transposed form in which it is recognized that it is an initial vibration that is carrying the system across the barrier. This would be expected to succeed since for the reverse reaction on surface II (the reverse of II is I) the adiabatic criterion is met and, from this standpoint at least, absolute rate theory is applicable in detail. If this transposition can be made, the most difficult case will become the case of the symmetrical energy barrier, considered by Wall and Porter,⁹ by Karplus, Porter, and Sharma,^{3b} and by Marcus.^{27,29}

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²⁵ R. A. Marcus, paper presented at the ACS Symposium, Atlantic City, September, 1968, and private communication.

²⁶ (a) J. O. Hirschfelder and E. Wigner, *J. Chem. Phys.* **7**, 616 (1939); (b) M. A. Eliason and J. O. Hirschfelder, *ibid.* **30**, 1426 (1959); (c) L. Hofacker, *Z. Naturforsch.* **18a**, 607 (1963); (d) R. A. Marcus, *J. Chem. Phys.* **41**, 2614 (1964); (e) **41**, 2624 (1964); (f) **43**, 1598 (1965).

²⁷ (a) R. A. Marcus, *J. Chem. Phys.* **45**, 2138 (1966); (b) **45**, 2630 (1966); (c) **45**, 4493 (1966); (d) **45**, 4500 (1966); (e) **46**, 959 (1967).

²⁸ There is some evidence, see ΔV in Fig. 5, that vibration makes a small additional contribution.

²⁹ The applicability of absolute rate theory to potential-energy surfaces with asymmetric barriers is being investigated in a detailed numerical study by M. Karplus and K. Morokuma (private communication).